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Dear Editors,

We are pleased to submit our manuscript, "*Passenger Willingness to Pay for Green Aviation Initiatives: A Global Meta-Analysis,*" for consideration for publication in *Transportation Research Part D: Transport and Environment.*

This study examines air passenger willingness to pay (WTP) for sustainable aviation fuel (SAF) surcharges and carbon offset programs through a meta-analysis of 17 international studies. We standardised WTP estimates and evaluated key geographic, demographic, and attitudinal predictors. Given the cost barriers associated with SAF adoption, understanding consumer willingness to absorb these costs is critical to the feasibility of large-scale aviation decarbonisation.

Our analysis identifies environmental concern as the most consistent predictor of WTP across studies. We also observe meaningful regional variation and a gap between stated WTP and actual participation in voluntary programs. These findings highlight the central role of environmental attitudes in shaping consumer willingness to support aviation decarbonisation initiatives, with potential implications for policy and engagement strategies.

We believe this manuscript is well-suited to *Transportation Research Part D: Transport and the Environment* given its focus on the intersection of transport systems and environmental outcomes. Our study contributes to the literature on sustainable aviation policy and consumer behaviour by providing a cross-study synthesis of factors influencing WTP and by standardising varied WTP estimates to a common unit, enabling more consistent comparison across studies.

This manuscript has not been published previously and is not under consideration by another journal. All authors have approved this submission. There are no conflicts of interest to declare. This study relies exclusively on previously published data and did not require ethics approval.

Thank you for your consideration. We would be happy to provide any additional information if needed.

Yours sincerely,

Fumnanya Jideonwor
Simon Fraser University

On behalf of all co-authors:

Katherine McKinnon, Simon Fraser University

Geoffrey Hong Kei Lee, Simon Fraser University

Jeffrey Song, Simon Fraser University

Suggested Reviewers

1. Stefano Carattini (Georgia State University)

2. Hanna K. L. Fageda (University of Barcelona)

3. Katrin Rehdanz (Kiel University)

Passenger Willingness to Pay for Green Aviation Initiatives: A Global Meta-Analysis

Katherine McKinnon
Fumnanya Jideonwor
Jeffrey Song
Geoffrey Hong Kei Lee

Department of Resource and Environmental Management
Simon Fraser University

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Katherine McKinnon, Fumnanya Jideonwor, Jeffrey Song, Geoffrey Hong Kei Lee

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Abstract

Aviation is a significant contributor to anthropogenic carbon emissions, particularly for long-distance travel where alternatives are limited. Despite advancements in aircraft technology, emissions from aviation remain a growing concern. This study explores different avenues for decarbonisation and assesses passenger receptiveness to the increased costs associated with emissions reductions (willingness to pay, or WTP). A meta-analysis of 17 international studies was conducted, standardising individual study findings to identify statistically significant patterns in WTP. Of the variables examined, environmental concern and awareness emerged as the most significant predictors of passenger WTP. Unlike fixed demographic factors such as income, gender, and education, environmental attitudes may be more amenable to change, suggesting their potential to shape support for more sustainable air travel.

Keywords: air travel; decarbonisation; offsets; SAF; willingness to pay; aviation; attitudes

1. Introduction

Aviation accounts for approximately 2.5% of global CO2 emissions, and its contribution is projected to increase as demand grows (Ritchie, 2024). Meanwhile, this sector is one of the most technically challenging to decarbonise; unlike land-based transport, where an electric revolution has already occurred, aviation faces fundamental limitations in battery technology that prevent a similar energy transition; even the most advanced batteries only produce 8-17% of the energy produced by jet fuel (Shafer et al., 2018). Therefore, alternatives such as sustainable aviation fuels (SAFs) may present a more immediately feasible pathway to decarbonisation.

SAFs are produced from waste oils, biomass, and other organic materials, rather than from petroleum or fossil-fuels (US Department of Energy, 2024). Before being used to produce fuel, plant matter removes carbon from the atmosphere through photosynthesis, and SAF production reuses organic material that would otherwise be wasted, such as woody byproduct and crop biomass (US Department of Energy, 2024). The IATA (2020) estimates that SAFs will produce 65% of emission reductions needed to achieve net-zero aviation by 2050, and compared to jet fuel, SAFs can reduce lifecycle emissions by up to 80% (Watson et al., 2024).

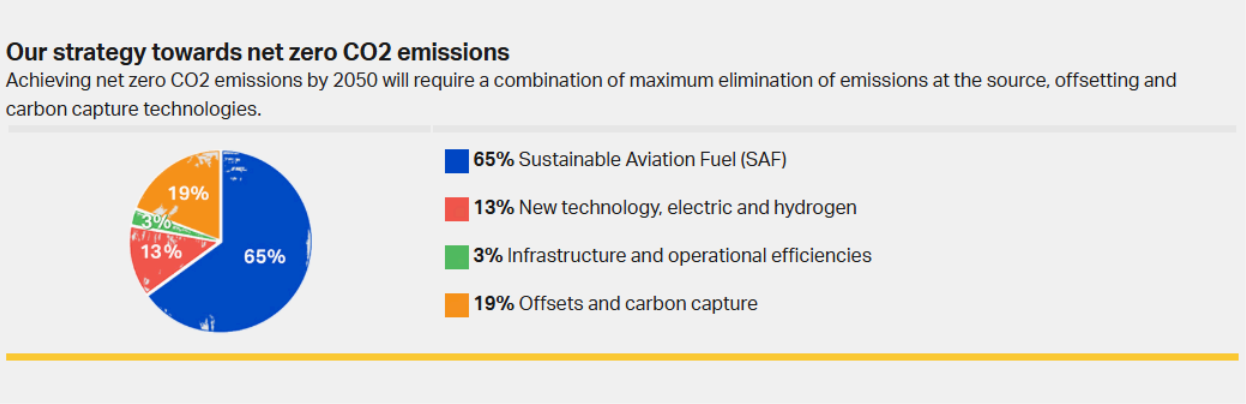


Figure 1: Pie chart showing the different strategies to pursue net zero emissions by 2050 (IATA, 2024).

Crucially, SAFs are also compatible with current airplanes and infrastructure, requiring no aircraft modification. At higher blend ratios, SAFs can effectively halve the emissions of a given flight (US Department of Energy, 2024). According to the US Department of Energy (2024), this alternative fuel can be found in more than 46 airports around the world, and it has already been used in over 360,000 flights.

While SAFs represent the most viable means of reducing aircraft emissions at scale, this technology is not without its shortcomings. The most significant of these is cost: SAFs currently cost between 120-700% more than conventional jet fuel (Watson et al., 2024). This is a major hurdle for industry adoption, where fuel costs already surpass 40% of operating expenses at a time when oil prices are already high (Dong et al., 2025). Airlines also operate on historically thin profit margins of 4-6%, guaranteeing that additional SAF costs will be passed on to travellers (Dong et al., 2025). Unless travellers are willing to bear these additional costs, voluntary SAF adoption is unlikely to reach meaningful scale without government subsidies or revolutionary industry change.

This study examines passenger WTP for aviation decarbonisation through a meta-analysis of 17 international studies, considering different geographic and demographic contexts, decarbonisation instrument types, and methodologies. The combined global sample enables broad, cross-study synthesis that no single study could provide– a particularly valuable approach for aviation, which is inherently global in nature.

This study addresses three research questions:

RQ1: What is the range of standardised passenger WTP for aviation decarbonisation across instrument types and geographic regions?

RQ2: Which sociodemographic and attitudinal variables most consistently predict passenger WTP across studies?

RQ3: Which aviation decarbonisation policy instruments are best positioned to translate passenger WTP into meaningful emissions reductions?

Prior to analysis, we hypothesised that:
H1: Environmental concern would be positively associated with WTP.

H2: Specific policy and fuel characteristics (such as program design, certification, and SAF blend ratio) would influence WTP.

H3: Price sensitivity would limit passenger WTP across studies.

2. Literature Review

2.1 Overview of Included Studies

The 17 studies examine passenger and traveller WTP across a range of aviation decarbonisation instruments, primarily voluntary carbon offsets, SAF surcharges, and carbon taxes, with several studies also examining information-based tools and mandatory pricing mechanisms. These studies collected data from Asian, European, Oceanian, North American, and global samples. Methodological approaches include contingent valuation, discrete choice experiments, conjoint analysis, and Likert-scale surveys. Studies varied considerably in how WTP was reported; some expressed values per tonne of CO_2, others as a percentage of ticket price or a fixed surcharge per flight, and several used Likert-scale attitude measures rather than monetary values (see Table 1).

Table 1. Overview of the 17 studies included in the meta-analysis

#	**Author (Year)**	**Country/Region**	**Instrument Type**	**Method**	**WTP Measure**	**Standardised WTP**
1	Lu & Shon (2012)	Taiwan	Carbon offset	Stated preference	$ per ticket	Yes
2	Jou & Chen (2015)	Taiwan	Carbon offset	Contingent valuation	NT$ per ticket	Yes
3	Chiambaretto et al. (2024)	Global	SAF surcharge	Conjoint analysis	EUR per ticket	Yes
4	Li et al. (2023)	Australia & NZ	Carbon offset & SAF	Discrete choice experiment	AUD per ticket	Yes
5	Sanguinetti & Amenta (2022)	USA	Carbon offset (informatio n nudge)	Stated preference	USD per tCO_2	Yes
6	Hong & Huh (2025)	South Korea	SAF surcharge	Contingent valuation	% airfare premium	Yes
7	Freitas et al. (2026)	Portugal	SAF surcharge	Path analysis / Likert scale	Likert scale	No
8	Hui et al. (2024)	Global	SAF surcharge & carbon offset	Stated preference	GBP per ticket	Yes

9	Rotaris et al. (2020)	Italy	Carbon offset	Discrete choice experiment	EUR per tCO_2	Yes
10	Rains et al. (2017)	USA	SAF surcharge (biofuel)	Likert scale survey	% of ticket price	No
11	Ritchie et al. (2021)	Australia	Carbon offset	Discrete choice experiment	AUD per offset	Yes
12	Brouwer et al. (2008)	Netherlands	Carbon offset	Contingent valuation	EUR per flight	Yes
13	Choi et al. (2018)	Australia	Carbon offset & carbon tax	Forced choice experiment	AUD per tCO_2	Yes
14	Blasch & Farsi (2014)	Switzerland	Carbon offset	Discrete choice experiment	USD per tCO_2	Yes
15	Rice et al. (2020)	USA	SAF surcharge	Likert scale survey	Likert scale	No
16	Wittmer & Mueller (2025)	Switzerland	SAF surcharge (corporate)	Adaptive conjoint	Likert scale	No
17	Choi & Ritchie (2014)	Australia	Carbon offset	Contingent valuation	AUD per tCO_2	Yes

2.2 Review of Key Findings

Regional patterns

Geographic variation is an important feature of this dataset, though care should be exercised when drawing conclusions about regional trends due to the uneven coverage and research design differences that will be further described in Section 2.3.

The European studies conducted by Brouwer et al. (2008), Rotaris et al. (2020), and Blasch & Farsi (2014) produced moderate but stable estimates of WTP. This stability may reflect the influence of existing carbon pricing infrastructure in European markets, where emissions trading systems and fuel taxes have created a familiar association between carbon costs and consumer prices, potentially anchoring passenger WTP to prevailing market rates. Significantly, the European estimates were also the most homogeneous across the region compared to any other geographic cluster.

The studies on Asian samples show a more dynamic trend. While earlier studies from Taiwan (Lu & Shon, 2012; Jou & Chen, 2015) had relatively lower WTP, the later study by Hong & Huh (2025) from South Korea had much higher values. This trend may reflect increasing attention to climate-related policies and carbon pricing mechanisms in East Asia. As a result, WTP in the region appears particularly sensitive to the evolving political context.

Australian studies demonstrated the greatest internal dispersion among all regions. Li et al. (2023) reported near-zero and negative WTP values, while Ritchie et al. (2021) and Choi & Ritchie (2014) found substantially higher estimates. This wide range suggests that WTP among Australian passengers is highly sensitive to instrument framing and survey design, which has important implications for how such preferences may be interpreted and applied in a policy context.

Sociodemographic factors

From the 17-study dataset, a subset of 13 studies examined sociodemographic WTP relationships in a comparable way. Five sociodemographic variables were examined as predictors of WTP: age, income, education level, gender, and environmental concern. Of these, environmental concern emerged as the most consistent predictor, with a positive association observed across all 13 studies (Hong & Huh, 2025; Choi et al., 2018, Lu & Wang, 2018, Hui et al., 2024). This pattern holds true regardless of region, instrument type, or methodological approach, suggesting that pro-environmental attitudes operate as a near-universal driver of WTP for green aviation initiatives.

The remaining four variables produced more uneven results. Income showed a positive relationship with WTP in the majority of studies where it was formally tested, particularly in

Asian and European contexts (Jou & Chen, 2015; Rotaris et al., 2020; Brouwer et al., 2008; Blasch & Farsi, 2014), though several studies, especially those based in North America and Oceania, either did not collect income data or found no significant effect. Education similarly trended positive in most studies that tested it, with higher-educated respondents generally expressing greater WTP, though exceptions exist: Blasch & Farsi (2014) found that holding an academic degree did not significantly affect stated preferences, and Choi & Ritchie (2014) found no significant education effect at all.

Age effects were more divided. Older respondents reported higher WTP in some studies (Jou & Chen, 2015), while younger respondents were found to be more willing to pay in others (Hong & Huh, 2025); Rotaris et al., 2020; Li et al., 2023), suggesting the relationship between age and WTP may be mediated by contextual factors such as flight frequency or familiarity with climate policy.

Gender was the least consistent variable: several studies found no significant gender effect (Jou & Chen, 2015; Blasch & Farsi, 2014; Chiambaretto et al., 2024), while others found women to have higher WTP than men (Choi & Ritchie, 2014; Rotaris et al., 2020). These inconsistencies across the literature point to the need for a structured cross-study synthesis, rather than relying on findings from any single study.

WTP estimates across instrument types

Across studies, WTP for aviation decarbonisation varied substantially by instrument type. The majority of studies (10/17) examined voluntary carbon offset programs, which reflects the commonality of this instrument, both in practice and in research. Among offset studies, WTP estimates ranged widely, from approximately \$8 USD/tCO_2e (Lu & Shon, 2012) to over \$272 USD/tCO_2e under ideal conditions (Ritchie et al., 2021). This suggests that passenger WTP is highly influenced by the way programs are designed and presented.

Studies that examined voluntary SAF surcharges (3/17) also reported varied WTP, ranging from 24 USD/tCO_2e (Hui et al., 2024) to \$260 USD/tCO_2e (Sanguinetti & Amenta, 2022). This range is comparable to WTP in offset studies, but has a higher lower value. Some studies suggest that passengers may be more willing to pay for SAF than offsets because the carbon emissions reductions are built directly into the flight, so the tool is perceived to be more effective (Chiambaretto et al., 2024; Rice et al., 2020).

Mandatory instrument WTP is less represented in the quantitative literature. Most studies that examined mandatory framing reported attitudes or percentage-based responses rather than monetary values (Rice et al., 2020; Rains et al., 2017; Freitas et al., 2026), making direct comparison with voluntary instrument WTP difficult. However, where monetary WTP for mandatory mechanisms was reported, attitudes toward carbon pricing significantly influenced those values– passengers who supported a carbon tax reported WTP more than three times higher

than non-supporters (Choi et al., 2018). This suggests that the political context in which mandatory instruments are introduced substantially affects WTP.

Policy and Governance

Beyond WTP values, several studies offer insights into the conditions under which voluntary decarbonisation instruments are more or less effective. A consistent finding across studies is that program design attributes — not just price alone — are the main determinants of passenger participation (Blasch & Farsi, 2014; Ritchie et al., 2021; Brouwer et al., 2008; Choi & Ritchie, 2014). Passengers consistently respond more positively to offset programs that are locally-sourced, government-certified, transparently reported, and administered by non-profit organisations (Blasch & Farsi, 2014; Ritchie et al., 2021; Brouwer et al., 2008). Conversely, programs that lack transparency, independent certification, or are administered by airlines or for-profit organisations have been associated with lower WTP and low consumer trust (Ritchie et al., 2021; Blasch & Farsi, 2014). Existing research also shows that passengers require evidence of substantial emissions reductions (at least 30–40%) to justify paying for voluntary SAF fees (Rice et al., 2020). Access to clear and credible information about aviation's environmental impact also increases WTP and participation (Sanguinetti & Amenta, 2022; Hui et al., 2024; Lu & Wang, 2018).

Despite these voluntary program design insights, however, actual uptake of these programs is very low: only 1–3% of passengers participate (Ritchie et al., 2021; IATA, 2020), which suggests that voluntary offset and SAF program improvements are insufficient (Berger et al., 2022). To create meaningful and measurable reductions in carbon emissions, aviation policy interventions may be necessary (Choi et al., 2018).

2.3 Limitations and gaps in existing research

The primary gap in the existing literature is geographic fragmentation– most studies are localised to specific regions, which is a significant limitation for an industry as globally interconnected as aviation. Findings from any single national or regional study are difficult to generalise across the international contexts that define air travel.

There is also no unit consensus across the existing literature. WTP has been expressed as a fixed surcharge per ticket, a percentage of fare, a value per tonne of CO_2, and as Likert-scale responses, with no easily comparable findings (Lu & Shon, 2012; Blasch & Farsi, 2014; Rice et al., 2020). Studies also vary widely in the instruments they examine: carbon offsets, SAF surcharges, carbon taxes, and voluntary versus mandatory mechanisms. These factors make direct cross-comparison difficult without standardisation.

Additionally, sociodemographic findings are typically reported within individual studies, making it difficult to identify clear patterns. For example, factors like income, education, gender,

age, and environmental concern have been examined in multiple studies, but the findings are inconsistent on a broader scale (Ritchie et al., 2021; Choi & Ritchie, 2014). Similarly, while individual studies have offered policy recommendations based on their own findings, a structured cross-study evaluation of aviation decarbonisation instruments against consistent criteria such as effectiveness, equity, and feasibility remains limited in the existing literature (Blasch & Farsi, 2014; Ritchie et al., 2021).

3. Methods

3.1 Study selection and inclusion criteria

This study initially aimed to conduct a quantitative analysis on the influence of different variables on WTP, so the original inclusion criteria required that papers contain coefficients that represented the impact of demographic, behavioural, and attitudinal variables, as well as a quantified measurement of WTP. However, since regression model types varied significantly across the literature, coefficients could not be compared reliably between studies. Therefore, the goal shifted to producing a descriptive analysis of these factors, and the original dataset was used to examine geographic patterns in the actual WTP estimates.

Studies were identified in late February and early March 2026 through the SFU Library search database, Google Scholar, and ScienceDirect, as well as backward citation searching of relevant articles. Search terms combined key terms (“willingness to pay” OR “WTP”) AND (“aviation” OR “air travel” OR “flight”) with additional terms related to instrument type (“sustainable aviation fuel” OR “SAF” OR “biofuel” OR “carbon offset” OR “carbon tax” OR “decarbonisation”). More specific search terms included (“regression” OR “coefficient”), (“logit” OR “logistic regression”), and (“environmental values” OR “environmental attitudes”).

Studies were excluded if they did not report a quantified measure of passenger WTP, defined as a monetary value per ticket, per tonne of CO2 emitted, or as a percentage of a specific ticket price or flight distance. Articles were also excluded if they did not examine the influence of at least one variable on WTP for decarbonisation, or if they focused on non-passenger populations (e.g. airline corporate decision-makers).

The 17 studies that met the inclusion criteria were not all used in the same way across the three analyses. Of the 17 studies, four reported WTP using Likert-scale or percentage-based measures only and could not be converted to a comparable monetary value (Rice et al., 2020; Freitas et al., 2026; Wittmer & Mueller, 2025; Rains et al., 2017). These four studies were therefore excluded from the quantitative WTP standardisation and regional ANOVA, which drew on the remaining 13 studies with standardised monetary WTP values. The sociodemographic analysis also drew primarily on these 13 studies, though Rice et al. (2020) was also referenced for a specific gender finding despite its exclusion from the quantitative WTP analysis. All 17

studies informed the policy evaluation framework in Section 3.5, alongside additional policy literature for context and depth of understanding.

3.2 WTP standardization procedure

Reported WTP values from selected studies were standardised to a common unit of USD per metric tonne of CO_2 equivalent (tCO_2e). This allowed for direct comparison across studies. The conversion process included four steps: emission factor application, radiative forcing adjustment, currency conversion, and inflation adjustment to 2023 values.

Per-passenger emissions were estimated using flight distance and DEFRA (2023) emission factors for short-haul and long-haul flights: 0.08993 and 0.07778 $kgCO_2$ per passenger-kilometre respectively. Where studies identified a specific route, great-circle distance was determined using Google Maps. Great-circle distance refers to the shortest path between two points on a globe, but in reality, aircraft travel less directly due to air traffic control routes, weather systems, and restricted airspace (DEFRA, 2023). A distance uplift factor of 8% was therefore applied to all great-circle distances to correct for this indirect routing

For studies that gave only a general route type without a specific origin or destination, a representative distance was assigned based on DEFRA (2023), which classifies flights as short-haul under 3700 km, and as long-haul over 3700 km. This assumption was necessary because several studies provided flight contexts without specifying exact routes. For studies that did not identify route distance or flight type, flight classification was deduced from contextual information; for example, a mean ticket price of $955 (Lu & Shon, 2012) likely implies a long-haul flight, and studies conducted on domestic flights in Australia were assigned a distance of 1400 km based on the average flight length in that context (BITRE, 2025).

A radiative forcing (RF) multiplier of 1.7 was then applied to per-passenger emission estimates. Radiative forcing refers to the total climate impact of aviation beyond CO_2 alone, since aircraft also produce nitrogen oxides, water vapour, and contrails when burning fuel at altitude, and these all contribute to climate change (Lee et al., 2021).The multiplier of 1.7 was selected because it represents a conservative, peer-reviewed midpoint estimate of the total climate impact of aviation, and has been widely included in aviation emission research to date (Lee et al., 2021). Chiambaretto et al. (2024) had already included this RF multiplier in its emissions estimate calculations, so no additional RF multiplier was applied here.

Some studies already expressed WTP per tCO_2 in local currency and did not require emission factor conversion (Rotaris et al., 2020; Choi et al., 2018; Blasch & Farsi, 2014; Choi & Ritchie, 2014). These estimates were simply converted to USD and adjusted for inflation.

Currency conversion to USD was performed using the average annual official exchange rates for each study's reference year (World Bank, 2023), and these values were adjusted to 2023 values using the US Consumer Price Index (US Bureau of Labor Statistics, 2023). The estimate from Sanguinetti & Amenta (2022) was already stated in USD/tCO_2e, and required only the adjustment for inflation.

3.3 ANOVA design

In order to determine whether geographic region was a statistically significant predictor of passenger WTP, a one-way ANOVA was conducted using the standardised WTP estimates obtained in Section 3.2. The use of one-way ANOVA is suitable for comparing mean values across more than two independent groups.

Based on the country or countries where the study took place, the studies under review were categorised into five regions, Asia, Europe, Oceania, Global and North America. The global category was intentionally retained because two studies (Chiambaretto et al., 2024; Hui et al, 2024) included participants from global sample groups without specifically reporting data by region. This made assignment of these studies into any other categories impossible based on the limited information provided.

However, North America was excluded from the ANOVA despite appearing in the sociodemographic analysis. Only two standardised data points were available for North America, both sourced from the same study (Sanguinetti & Amenta, 2022). Two data points were insufficient to generate a reliable group mean consistent with ANOVA assumptions of adequate group size. The final ANOVA was therefore performed across only four groups: Asia (n=6), Europe (n=20), Global (n=9), and Oceania (n=31).

3.4 Sociodemographic variable coding

Because the included studies reported sociodemographic variables in different ways– some providing regression coefficients, others qualitative narrative findings, and several omitting certain variables entirely— a standardised coding framework was necessary to enable cross-study comparison. Each of the five variables (age, income, education, gender, and environmental concern) was coded into one of four directional categories: *Positive* (higher value associated with positive WTP), *Negative* (negative value associated with lower WTP), *Mixed* (inconsistent or indeterminate effects reported within the study), or *Not Reported/Not Significant* (data either not collected, not analysed, or found to have no statistically significant effect).

Coding decisions were made on the basis of the primary findings reported in each study, with reference to the direction and statistical significance of the regression coefficients where available. Where a study reported significance at the $p < 0.05$ level, the effect was coded as *Positive* or *Negative* accordingly. Where significance was not proven, or where it varied across

sub-samples, the variable was coded as *Mixed*. Where a variable was collected but not included in any regression model, or explicitly described as insignificant without further elaboration, it was coded as *Not Reported*. This approach prioritised consistency and replicability over granularity, acknowledging that the underlying studies used different instruments, scales, and analytical models that cannot be directly harmonised beyond directional coding.

3.5 Policy evaluation framework

The policy analysis was based on the five-criteria framework developed by Kraft & Furlong (2015). Eight aviation decarbonisation instruments were evaluated based on their effectiveness, efficiency, equity, acceptability, and feasibility. These instruments were grouped into three categories: voluntary, regulatory, and information/governance tools. They were scored on a scale of 1 to 3, where 1=low, 2=moderate, and 3=high performance in that criterion. The analysis was based on evidence from the included studies and from existing policy literature.

The five criteria were defined as follows: effectiveness captures the degree to which an instrument produces measurable and meaningful emissions reductions in practice. Efficiency refers to the cost-effectiveness of a policy, especially relative to emissions reduced. Equity considers how proportionally and fairly the financial or environmental impacts of a policy were distributed. Acceptability assesses the degree of resistance a policy would likely provoke among passengers, industry, and policymakers. Feasibility refers to the severity of barriers to actually implementing a policy or tool.

The decision rules outlined in Table 2 were applied consistently across all instruments:

Table 2: Policy Analysis Decision Rules

Criterion	**1 (Low)**	**2 (Moderate)**	**3 (High)**
Effectiveness	- Produce no direct or effective emissions reduction - may have low real-world uptake	- Produce partial or indirect emissions reductions	- Produce direct, measurable, significant emissions reductions
Efficiency	- Prohibitively high cost per unit of emissions reduced	- Moderate implementation, verification, or enforcement costs	- Meaningful impact achievable at low cost within existing infrastructure
Equity	- Failure to distribute financial or environmental costs in	- Financial or environmental costs are shared more	- The greatest burden falls to the highest emitters

	proportion to responsibility for emissions	broadly/uniformly - No differentiation by flight frequency or income	- May place no additional financial burden on passengers
Acceptability	- Strong opposition likely from passengers, industry, or policymakers	- Mixed reception across stakeholder groups	- Broadly acceptable to public, industry, and government
Feasibility	- Major structural, political, or regulatory barriers to implementation	- Requires new legislation, international coordination, or significant administrative capacity	- Necessary infrastructure already exists; no major new mechanisms required.

4. Results

4.1 Standardized WTP Values

Table 1 illustrates the standardised WTP estimates for all 13 studies included in the quantitative analysis. The 68 data points from these studies showed that standardised WTP ranged from –$4.20 to $272.31 USD/$tCO_2e$, with an overall median of $24.44 and a mean of $44.51 The overall median was $24.44 and the mean was $44.51 At the study level, the median WTP ranged from $1.80 (Li et al., 2023) to $225.68 (Sanguinetti & Amenta, 2022), indicating significant variation across different geographies and instrument types.

4.2 Regional ANOVA results

A one-way ANOVA was conducted on the 68 standardized WTP data points from the four regions with adequate samples: Asia (n = 6), Europe (n = 20), Global (n = 9), and Oceania (n = 31). As previously discussed, North America was omitted from the ANOVA due to an insufficient number of data points (n = 2), both derived from a singular study (Sanguinetti & Amenta, 2022).

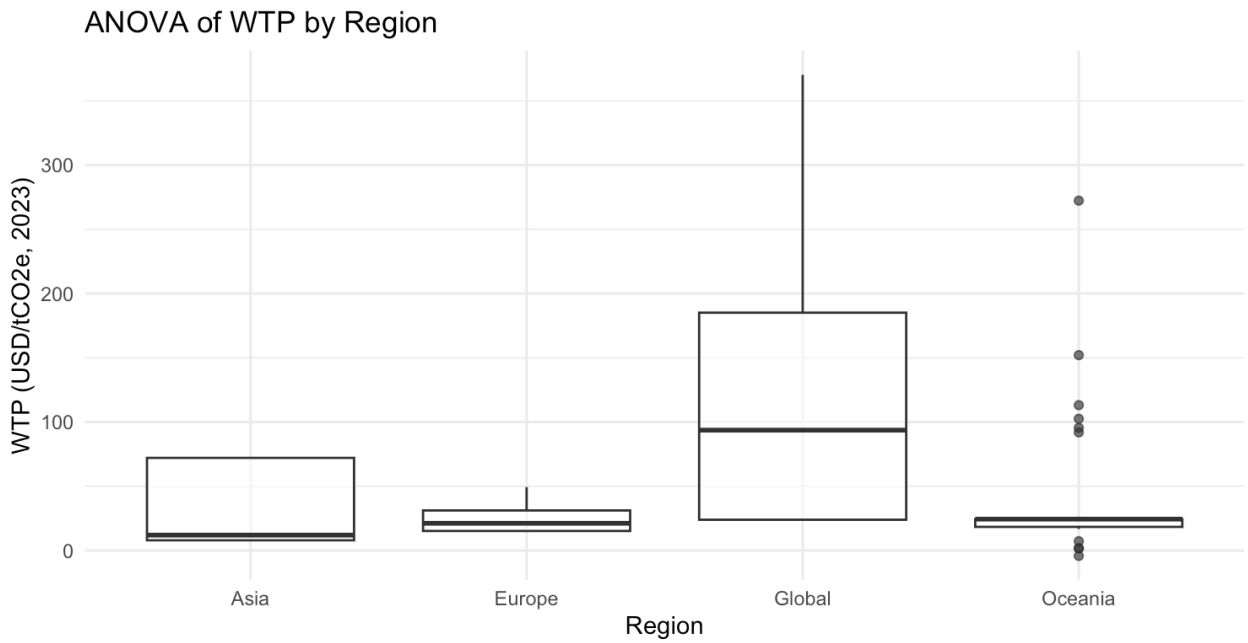


Figure 2: *ANOVA of standardized WTP by region (2023 USD/tCO_2e). Boxes show interquartile range (IQR), whiskers extend to 1.5. IQR dots are outliers*

The ANOVA returned a statistically significant result: $F(3, 72) = 7.27$, $p = 2.52 \times 10^{-4}$, indicating that regional group membership is a significant predictor of WTP. Figure 2 presents the distribution of standardized WTP values by region.

Table 3: Summary statistics of standardized WTP by region (2023 USD/tCO_2e)

Region	n	Mean (USD/tCO_2e)	Median	Min	Max	Std Dev
Oceania	31	42.24	24.44	-4.2	272.31	55.94
Europe	20	23.71	21.2	15.24	49.38	9.79
Global	9	53.68	23.98	23.98	119.23	38.62
Asia	6	51.4	72.11	7.98	72.11	32.1
North America*	2	225.68	225.68	191.36	260	48.54

*North America excluded from ANOVA due to insufficient sample size (n = 2, single study). Reported here for descriptive completeness only.

The Global group had the highest mean WTP at \$53.68/tCO_2e, but the wide range of emissions scenarios in Chiambaretto et al. (2024) produced considerable spread (median = \$23.98, range = \$23.98–\$119.23). Asia recorded a mean of \$51.40 and a median of \$72.11, with the majority of data points drawn from Hong & Huh (2025); Lu & Shon (2012) and Jou & Chen (2015) contributed lower values that extended the range downward. Oceania recorded a mean of \$42.24 but a median of only \$24.44, reflecting the disproportionate influence of the high-end estimates in Ritchie et al. (2021) relative to the near-zero values reported by Li et al. (2023). Europe recorded the lowest variance of any group (mean = \$23.71, median = \$21.20, standard deviation = \$9.79), with estimates drawn from Brouwer et al. (2008), Rotaris et al. (2020), and Blasch & Farsi (2014). The significant F-statistic, combined with Europe's notably lower and more similar estimates, indicates that regional context meaningfully shapes passenger WTP. This pattern will be examined further in Section 5.1.

4.3 Sociodemographic patterns

Environmental concern was coded as *Positive* in all 13 studies in which it was examined. This was the only variable to return a consistently positive result across the dataset. No study recorded a *Negative*, *Mixed*, or *Not Reported* result for this variable.

Income was coded as *Positive* in five studies (Jou & Chen, 2015; Brouwer et al., 2008; Blasch & Farsi, 2014; Rotaris et al., 2020; Chiambaretto et al., 2024), and *Not Reported/Not*

Significant in three (Choi & Ritchie, 2014; Choi et al., 2018; Hui et al., 2024). No study returned a *Negative* income effect. Five studies did not formally test income.

Education was coded as *Positive* in three studies (Jou & Chen, 2015; Rotaris et al., 2020; Choi et al., 2018), *Mixed* in two (Blasch & Farsi, 2014; Li et al., 2023), and *Not Reported/Not Significant* in two (Choi & Ritchie, 2014; Hui et al., 2024). No study returned a *Negative* education effect.

Age showed the most divided results of the four demographic variables. Three studies coded age as *Positive*. In this context, *Positive* meant that older passengers expressed higher WTP (Jou & Chen, 2015). Three studies coded it as *Negative*, indicating that younger passengers expressed higher WTP (Hong & Huh, 2025; Rotaris et al., 2020; Li et al., 2023). No consistent pattern could be established across regions.

Gender was the least informative variable in the dataset. Seven studies gave a *Not Reported/Not Significant* result for gender (Jou & Chen, 2015; Blasch & Farsi, 2014; Brouwer et al., 2008; Choi et al., 2018; Chiambaretto et al., 2024; Hong & Huh, 2025; Hui et al., 2024). Among the studies that did find a gender effect, results were coded as *Mixed* — Choi & Ritchie (2014) and Rotaris et al. (2020) found women expressed higher WTP, while Rice et al. (2020) found that any gender effect for domestic flights disappeared in the international context.

4.4 Policy instrument scoring

Table 4 below illustrates the scores assigned to each of the eight decarbonisation policies across the five criteria, along with total scores for each policy out of 15. Scoring rules are outlined in Table 2 of the Methods section.

Table 4: Decarbonisation Policy Analysis Results

Policy	Effectiveness	Efficiency	Equity	Acceptability	Feasibility	Total Score (/15)
Voluntary Instruments						
Voluntary Carbon Offsets	1	2	1	3	3	**10**
Voluntary SAF Surcharge	1	1	1	3	3	**9**
Regulatory Instruments						
Mandatory SAF Blending Targets	2	2	2	2	3	**11**
Ring-Fenced Aviation Carbon Levy	3	3	2	2	2	**12**
Frequent Flyer Levy	3	2	3	1	1	**10**
Information/Governance Instruments						
Mandatory Carbon Labelling at Point of Search	2	3	3	3	3	**14**
Government Certification Standards for SAF & Offsets	2	3	2	3	3	**13**
Mandatory Corporate Scope 3 Emissions Reporting	2	3	2	2	3	**12**

Score Interpretation:		
Low (1)	**Moderate (2)**	**High (3)**

Overall, policies involving voluntary instruments received the lowest scores. The Voluntary SAF Surcharge scored 9/15, with low scores on effectiveness, efficiency, and equity. Voluntary Carbon Offsets scored similarly low on effectiveness and equity, but higher scores on acceptability and feasibility brought its total to 10/15.

Regulatory instruments performed marginally better. Within this policy category, the Ring-Fenced Aviation Carbon Levy scored highest at 12/15, through its high effectiveness and efficiency rankings.The Mandatory SAF Blending Targets scored moderately across most criteria and highly on feasibility for a total score of 11/15. The Frequent Flyer Levy also ranked poorly overall, with a total score of 10/15: it performed highly in effectiveness and equity, but low on acceptability and feasibility.

Information and governance tools scored highest overall. Mandatory Corporate Scope 3 Emissions Reporting and Government Certification Standards for SAF and Offsets scored 12/15 and 13/15 respectively, with consistently high scores on efficiency and feasibility. Mandatory Carbon Labelling at Point of Search received the highest total score of any instrument at 14/15, scoring high across four of the five criteria, with only effectiveness ranked as moderate.

5. Discussion

5.1 Regional variation

The findings of the ANOVA, presented in Section 4.2, demonstrate that geographic region is a statistically significant determinant of passenger WTP ($F(3, 72) = 7.27$, $p = 2.52 \times 10^{-4}$). This interesting variance is observable in standardised WTP values, which reflect not only

differences in passenger attitudes and values, but also reflect differences in the economic environments and aviation policy contexts of each region.

The lower WTP in Europe– with a mean of \$23.71/tCO_2e and the lowest standard deviation of all regions– is surprising given the role Europe often plays in progressive climate policy. Yet the findings align with prior literature on the influence of existing carbon pricing systems on WTP. European consumers, especially those in the Netherlands (Brouwer et al., 2008) and Italy (Rotaris et al., 2020), operate within regulatory regimes that feature cap-and-trade systems and fuel taxation. Where carbon costs are already factored into airline tickets through existing regulatory mechanisms, passengers may see little need for additional voluntary contributions, or instead base their stated WTP on the low market carbon price levels. Similarly, Blasch & Farsi (2014) found that in Switzerland, high environmental awareness did not translate into higher voluntary WTP.

Asia's higher mean WTP (\$51.40) reflects an interesting contrast. South Korea has been very vocal about its carbon-neutral goals and has been actively trading its carbon emissions (Hong & Huh), 2025. However, compared to Europe, aviation-specific carbon initiatives for consumers are relatively new in Asia. Where passengers have heightened level of consciousness of environmental issues and little experience in paying for carbon emissions on flights, stated WTP values may be higher simply due to the novelty of the idea.

The high variance in Oceania is by far the most instructive regional finding. Low estimates in Li et al. (2023) contrast with high estimates in Ritchie et al. (2021), which illustrates how Australian WTP varies considerably depending on the framing of the instrument and the emissions scenario used, rather than based on an underlying preference. This suggests that instrument design and communication may be as influential as underlying attitudes in determining WTP.

Overall, while environmental attitudes are undoubtedly important predictors of WTP for green aviation, they do not determine it in isolation but rather in conjunction with the specific economic and regulatory context. This has direct policy implications: instruments designed for a particular region cannot be transplanted elsewhere without accounting for existing carbon pricing structures and passenger familiarity with carbon costs.

5.2 Sociodemographic Predictors

The sociodemographic patterns identified in Section 4.3 present a nuanced picture that both reinforces and complicates assumptions common in the aviation sustainability literature. The most striking finding is the dominance of environmental concern as a predictor of WTP. Across all 13 studies, without exception, higher environmental concern, pro-environmental attitudes, or belief in the effectiveness of green aviation instruments was associated with greater willingness to pay (see Section 4.3). This is consistent with established theories of

pro-environmental behaviour, including the Theory of Planned Behaviour applied by Hui et al. (2024), which positions attitudes as the primary antecedent of behavioural intention. Chiambaretto et al. (2024) quantified this gap most clearly: travellers with high pro-environmental behaviour expressed a WTP more than five times greater than their low-behaviour counterparts, highlighting how deeply attitudes divide the passenger population.

However, this finding carries an important implication that goes beyond individual variation. If WTP is so contingent on pre-existing environmental attitudes, then voluntary mechanisms (which rely on passengers choosing to opt in) will mostly only draw on the small subset of travellers who are already environmentally motivated. This reflects a structural selection problem rather than consumer apathy, and partially explains the well-documented gap between stated WTP and actual participation that will be discussed in section 5.3.

This connects to what the literature has described as the frequent flyer paradox. High-frequency travellers, those who generate disproportionately large per-capita aviation emissions, have been found to be among the least likely to pay for offsets or SAF surcharges voluntarily (Gössling & Humpe, 2020; Choi et al., 2018; Lu & Shon, 2012). Choi et al. (2018) found that flight frequency had no significant positive effect on WTP, and in some model specifications was associated with lower offset valuation. This pattern suggests that frequency of flying may erode rather than heighten a sense of personal emissions responsibility, possibly reflecting the normalisation of high-carbon consumption among frequent travellers (Gosseling & Humpe, 2020; Choi et al., 2018). This paradox is an important consideration for policy: those with the greatest individual impact on aviation emissions are also those least likely to respond to voluntary pricing instruments, meaning that voluntary schemes will tend to redistribute responsibility toward low-frequency, lower-emission passengers who are already more environmentally engaged. Overall, these findings challenge the premise that voluntary payment options are an adequate response to aviation's climate impact.

The income findings further complicate the equity dimensions of voluntary instruments. Where income effects were significant (especially in European and Asian samples), higher-income respondents expressed greater WTP. If the passengers most willing to pay are also the wealthiest, voluntary schemes risk becoming a green premium rather than an effective emissions reduction mechanism. This does not mean that income-based differentiation in mandatory instruments is straightforward, but it does suggest that the distributional effects of different policy designs deserve more explicit attention than the existing literature has provided.

Generally, the sociodemographic evidence reviewed highlights the necessity of mandatory or regulatory instruments that do not depend on voluntary pro-environmental motivation to achieve meaningful coverage across the passenger population.

5.3 WTP vs actual behaviour gap

The most relevant policy insight from this research is not necessarily the amount that people are willing to pay, but rather the difference between their stated willingness to pay and actual behaviour. In voluntary carbon offset programs, only an average of 1-3% of passengers participate (Gössling & Humpe, 2020). The gap between a median standardized WTP of $\$24.44/tCO_2e$ is incredibly far removed from reality: near-zero participation. This discrepancy cannot be explained by consumer apathy alone, and justifies a more structural explanation.

The elicitation of survey-based WTP figures takes place within a structural context that is distinct from the context in which passengers make their actual buying decisions. In the survey context, there is increased emphasis on payment for carbon offsetting, which makes the activity morally salient and public. However, when booking the flight, there are various other considerations (seat fees, luggage fees, and even competitive pricing by other airline companies) that make the WTP of $\$24/tCO_2e$ unrealistic.

The difference is reflective of the relationship between willingness-to-pay and willingness-to-accept. Willingness-to-pay refers to how much an individual would voluntarily offer, while willingness-to-accept refers to how much they will tolerate being charged before abandoning a transaction. According to studies in behavioral economics, the estimate of willingness-to-accept estimates consistently exceed willingness-to-pay; people tend to value loss more than they value equivalent gains (Kahneman & Tversky, 1979). Accordingly, an automatic charge instead of a voluntary SAF surcharge may result in resistance from passengers who expressed high WTP in survey contexts, not because their environmental beliefs have changed, but because of loss aversion triggered by a non-voluntary framing.

Therefore, the behaviour gap is a failure of governance, not of consumerism. If voluntary schemes fail to reach more than 1–3% of people no matter how they are structured or promoted, the limit to voluntary adoption has effectively been reached. The other 97–99% of flyers will not become more likely to adopt these policies through improvements in messaging or labelling alone. Closing the behavioural gap requires non-voluntary mechanisms that ensure passengers act on their stated WTP without requiring that they actively choose to opt-in. Mandatory SAF blending quotas, automatic fare surcharges (with opt-out provisions), and sector-wide carbon pricing have the potential to effect real change here.

5.4 Further policy considerations

As established in Section 5.3, the behaviour gap reflects a governance failure rather than a consumer one, and the results of the policy analysis make that clear. There is a fundamental trade-off in aviation decarbonisation policy: the instruments most likely to produce meaningful

emissions reductions are also the least politically and socially acceptable. Regulatory instruments received the highest effectiveness scores but also the lowest acceptability and feasibility scores, while voluntary instruments showed the reverse pattern. This trade-off cannot be resolved, but it can be managed.

This does not mean that voluntary instruments should be abandoned altogether. The evidence suggests they would be more effective with support from two complementary approaches: information tools that activate the environmental concern shown to drive WTP (Lu & Wang, 2018), and governance tools that improve the transparency, credibility, and quality of the options available to passengers (Blasch & Farsi, 2014; Ritchie et al., 2021).

The case for information tools begins the identification of environmental concern as the strongest predictor of WTP across studies(see Section 4.3), yet most passengers have low awareness of aviation's actual emissions contribution. Without awareness of the problem, concern cannot develop, and voluntary action is unlikely to follow (Lu & Wang, 2018; Hui et al., 2024). Carbon labelling at the point of search (Score: 14/15) addresses this directly: by making emissions visible while passengers are actively comparing flight options, it increases awareness and has been shown to influence choice toward lower-emission alternatives (Sanguinetti & Amenta, 2022; Hui et al., 2024). To be most effective, labelling should be mandatory and standardised across all airlines and booking platforms, ensuring all passengers have access to reliable, comparable emissions information regardless of where they book (Sanguinetti & Amenta, 2022; Ritchie et al., 2021).

However, even among aware and concerned passengers, low trust in offset program quality and limited transparency remain key barriers to voluntary participation (Blasch & Farsi, 2014; Ritchie et al., 2021). Research shows that passengers respond more positively to programs that are locally sourced, government certified, transparently reported, and administered by non-profit organisations (Ritchie et al., 2021; Blasch & Farsi, 2014; Brouwer et al., 2008). Passengers also require evidence of substantial emissions reductions — a threshold of at least 30–40% has been identified as necessary to generate positive WTP for SAF instruments (Rice et al., 2020). Government certification standards for SAF and offsets (Score: 13/15) address these preferences directly, and must be mandatory rather than voluntary to avoid creating a two-tier market that undermines the credibility certification is intended to provide (Blasch & Farsi, 2014). Crucially, both mandatory labelling and certification place compliance obligations on airlines and booking platforms rather than on passengers. This imposes no direct financial burden to them and therefore avoids the acceptability trade-off that characterises mandatory financial instruments (Sanguinetti & Amenta, 2022; Ritchie et al., 2021; Blasch & Farsi, 2014).

Ultimately, the evidence points not to a single instrument but to a sequence. Information and governance tools should come first, building the awareness, trust, and political conditions under which stronger regulatory mechanisms become viable. The trade-off between effective and

acceptable policy cannot be eliminated, but it can be managed through careful sequencing and design.

Even with stronger information and governance tools in place, however, the equity dimensions of voluntary instruments remain a concern. As established in Section 5.2, where higher-income passengers are consistently more willing to pay, voluntary schemes risk functioning as green premium consumption rather than systematic decarbonisation — redistributing responsibility toward those least responsible for aviation emissions. This reinforces the case for mandatory instruments designed with equity in mind. While information and governance tools represent the most immediately viable path forward, this analysis does not rule out mandatory financial instruments — it highlights that their success depends on addressing the concerns that make them unpopular (Ritchie et al., 2021; Blasch & Farsi, 2014). A ring-fenced aviation carbon levy represents the most structurally effective long-term mechanism (CE Delft, 2024; NEF, 2024), but its viability requires careful design: ring-fencing revenue for aviation decarbonisation improves public support and transparency (Ritchie et al., 2021), equity concerns can be addressed through rebates or travel credits for lower-income passengers (Blasch & Farsi, 2014), and feasibility is supported by existing aviation tax infrastructure and regulatory precedent in the EU and UK (European Union, 2023; United Kingdom Government, 2024).

Ultimately, the evidence points not to a single instrument but to a combination of them. Information and governance tools should come first, building the awareness, trust, and political conditions under which stronger regulatory mechanisms become viable. The trade-off between effective and acceptable policy cannot be eliminated, but it can be managed through careful sequencing and design.

6. Conclusion

Aviation is one of the final sectors to face meaningful decarbonisation pressure, with the arrival of sustainable aviation fuels, carbon offsets, and carbon taxes. Due to the increased costs associated with reducing emissions in flight, not everyone is willing to pay for green aviation. This meta-analysis demonstrated that the discrepancy between those who are and are not willing to pay mostly boils down to environmental awareness and concern– the attribute most consistently associated with higher WTP across all 13 quantitative studies. It is also important to reiterate that the more effectively a tool addresses emissions, the less publicly acceptable it tends to be, since mandatory implementation forces everyone to pay more regardless of preference. However, the most socially acceptable mechanisms are often the least effective, with voluntary programs seeing only 1–3% actual participation: this exposes a persistent rift between stated support and actual action (Gössling & Humpe, 2020; Ritchie et al., 2021; IATA, 2020), a discrepancy that has been highlighted through this review. This study has also demonstrated that meaningful WTP foundation exists among a significant proportion of the flying public; the

challenge lies in translating that WTP into actual emissions reductions through effective governance.

Our recommendation is that policy development follow a combined approach: first, prioritise information and governance tools, including mandatory carbon labelling at the point of flight search and government certification standards for SAF and offset programs. This would serve to build awareness, trust, and transparency, which would then foster the socio-political conditions necessary for stronger regulatory mechanisms (such as a ring-fenced aviation carbon levy). No single instrument will resolve the tension between effective and acceptable policy, and the limitations outlined in Section 2.3 all still apply. Future research should focus on practical piloting of different policy combinations to identify the most effective sequencing across different regional and regulatory contexts. Aviation decarbonisation is an achievable goal, but it will require policy ambition to match the scale of the challenge.

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